

Subtract First Digit

Statement

Start with a positive integer n .

Repeatedly, take the first digit of your current number and subtract it from the number.

Eventually, you'll reach zero. Count the steps until that happens.

Report that count modulo $10^9 + 7$ (to keep the output small).

Input format

The first line of each input file contains the number t of test cases. The specified number of test cases follows, one after another.

Each test case consists of a single line containing one value of n .

Output format

For each test case output a single line with one integer: the value $(x \bmod 1\,000\,000\,007)$, where x is the number of steps needed to get from n to 0.

Subproblem S1 (30 points, public)

Input file: [S1.in \(/static/qual2026/S1.a1a47fac11102711bc8eafbfaa7ab10c.in\)](/static/qual2026/S1.a1a47fac11102711bc8eafbfaa7ab10c.in).

Constraints: $t = 100$ and each n has at most 7 digits.

Subproblem S2 (60 points, public)

Input file: [S2.in \(/static/qual2026/S2.524941def3a36ddda744e8cd78477555.in\)](/static/qual2026/S2.524941def3a36ddda744e8cd78477555.in).

Constraints: $t = 100$ and each n has at most 18 digits.

Subproblem S3 (10 points, secret)

Input file: [S3.in \(/static/qual2026/S3.11ccfefbfff03f6ea451ce21f868a677b.in\)](/static/qual2026/S3.11ccfefbfff03f6ea451ce21f868a677b.in).

Constraints: $t = 100$ and each n has exactly 20 000 digits.

Examples

input	output
4 8 17 37 1234567	1 9 18 548891

The example input contains $t = 4$ test cases: $n = 8$, $n = 17$, $n = 37$, and $n = 1\,234\,567$.

For $n = 8$ we subtract 8 from 8 and get 0, so we are done in a single step.

For $n = 17$ the process looks as follows: $17 \rightarrow 16 \rightarrow 15 \rightarrow 14 \rightarrow 13 \rightarrow 12 \rightarrow 11 \rightarrow 10 \rightarrow 9 \rightarrow 0$.

For $n = 37$ the following will happen: $37 \rightarrow 34 \rightarrow 31 \rightarrow 28 \rightarrow 26 \rightarrow 24 \rightarrow 22 \rightarrow 20 \rightarrow 18 \rightarrow 17 \rightarrow 16 \rightarrow 15 \rightarrow 14 \rightarrow 13 \rightarrow 12 \rightarrow 11 \rightarrow 10 \rightarrow 9 \rightarrow 0$.

input	output
1 4567890123	289635647

The second example contains a single test case with $n = 4\,567\,890\,123$. Note that this n is bigger than the constraint for subproblem S1, but it could appear in subproblem S2.

For this n , the total number of steps is $x = 2\,289\,635\,661$, so the correct output is the remainder this x gives when divided by $10^9 + 7$.